

Home Assignment

- 1) Check for TL:DR ,ELI -15 in prompting and see the difference in output
- 2) Using chain of thoughts
 - a)Generate automation framework architecture
 - b)Generate Test strategy document

Answers:

- 1) Check for TL:DR ,ELI -15 in prompting and see the difference in output

Answer:TL:DR https://chatgpt.com/s/t_69ffe3105ce481919406680821bc828b

Answer : ELI-20: https://chatgpt.com/s/t_69ffe71b0df48191b898d0d4f3632290

- 2-A) COT(Generate automation framework architecture)

URL- https://chatgpt.com/s/t_69ffdeffbdb881919c45e8813f9495cb

- 2-B) COT Test Strategy document:

Transportation Quote and Booking Process- Test Strategy Document

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1. Project Overview

This document defines the Test Strategy for the Quote and Booking process in a Multi-modal Transportation platform covering Road, Air, and Ocean shipment workflows. The platform is based on Microservices Architecture and supports end-to-end customer shipment booking capabilities.

2. Business Objectives

- Ensure accurate shipment booking workflows
- Validate transportation quote generation
- Ensure reliable shipment confirmation
- Validate carrier integration and booking acknowledgements
- Ensure high system availability and operational stability

3. Testing Scope

In Scope:

- Quote generation process
- Booking creation and confirmation
- Rate calculation validation
- Customer and shipment information validation
- Carrier API integrations
- Database validations
- UI and API testing
- Performance and security testing

Out of Scope:

- Billing and invoicing
- Warehouse operations
- Driver mobile applications

4. Test Types

- Functional Testing
- API Testing
- Integration Testing
- Database Testing
- End-to-End Testing
- Regression Testing
- Smoke Testing
- Performance Testing
- Security Testing
- UAT Support

5. Application Architecture

The application follows a Microservices Architecture where services communicate through REST APIs and event-driven integrations.

Key services include:

- Quote Service
- Booking Service
- Carrier Integration Service
- Pricing Engine
- Notification Service
- Authentication Service

6. Test Environment Strategy

Environment Types:

- SIT Environment
- QA Environment
- Pre-Production Environment

Required Components:

- API Gateway
- Microservices Containers
- Database Servers
- Message Queues
- Third-party Carrier Sandbox APIs

7. Automation Strategy

Automation Tool: Playwright with JavaScript

Automation Coverage:

- UI Automation
- API Automation
- Cross-browser testing
- Data-driven testing
- Regression suite execution

Framework Features:

- BDD implementation
- Reusable page objects
- CI/CD integration
- Reporting and screenshots

8. Test Data Management

- Shipment master data
- Customer profiles
- Carrier test accounts
- Rate card datasets
- Multi-country shipment data
- Negative test datasets

9. Entry and Exit Criteria

Entry Criteria:

- Stable build deployed
- Test environment available
- Test data prepared
- APIs accessible

Exit Criteria:

- Critical defects closed
- Regression suite passed
- Test coverage achieved
- Stakeholder sign-off completed

10. Defect Management

Defect Lifecycle:

New → Assigned → In Progress → Fixed → Retest → Closed

Severity Levels:

- Critical
- High
- Medium
- Low

11. Risk Management

Potential Risks:

- Carrier API instability
- Incorrect freight rates
- Booking failures
- Environment downtime
- Delayed third-party integrations

Mitigation:

- Mock service implementation
- Retry validation
- Continuous monitoring
- Early integration testing

12. Deliverables

- Test Strategy Document
- Test Plan
- Test Scenarios
- Test Cases
- Automation Scripts
- Defect Reports
- Test Summary Report

13. Roles and Responsibilities

Test Manager:

- Overall test governance
- Risk management
- Stakeholder communication

Automation Engineer:

- Automation framework development
- Script maintenance
- CI/CD execution

QA Analyst:

- Test execution
- Defect reporting
- Validation activities

14. Tools and Technologies

- Playwright
- JavaScript
- Jenkins
- GitHub
- Postman
- SQL
- JIRA
- Docker

15. Conclusion

This strategy ensures robust validation of the transportation Quote and Booking process with focus on booking reliability, system stability, API integration quality, and scalable automation execution.